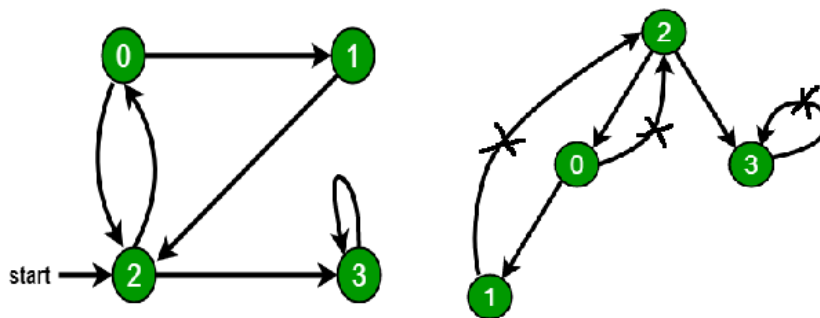


Depth First Search or DFS for a Graph

Depth First Traversal (or Search) for a graph is similar to Depth First Traversal of a tree. The only catch here is, unlike trees, graphs may contain cycles, so we may come to the same node again. To avoid processing a node more than once, we use a boolean visited array.

For example, in the following graph, we start traversal from vertex 2. When we come to vertex 0, we look for all adjacent vertices of it. 2 is also an adjacent vertex of 0. If we don't mark visited vertices, then 2 will be processed again and it will become a non-terminating process. A Depth First Traversal of the following graph is 2, 0, 1, 3.



See [this post](#) for all applications of Depth First Traversal.

Following are implementations of simple Depth First Traversal. The C++ implementation uses [adjacency list representation](#) of graphs. [STL](#)'s [list container](#) is used to store lists of adjacent nodes.